Lecture 22

Chern classes and the Dirac monopole

Objectives. *A connection can change while its bundle stays fixed. We extract a cohomology class from its curvature that survives this change. On a sphere its integral equals the transition winding and is therefore an integer. For a charged particle in a monopole field, this integer gives Dirac's quantization condition. The tangent sphere supplies a second example through Gauss–Bonnet.*

By the end of this lecture you will be able to:

- Prove that a line bundle's curvature class is independent of its connection.
- Recover its sphere Chern number from the oriented equatorial transition.
- Derive Dirac quantization from the two smooth monopole potentials.
- Relate the tangent sphere and the degree-n line to their Chern numbers.

Unit 22A Curvature classes and monopole flux

22.1 The curvature class of a line bundle

Let $E \to M$ be a complex line bundle with a Hermitian fibre metric and a unitary connection. In a unitary frame e_i, the connection's imaginary potential A_i has curvature $F_i = \mathrm{d}A_i$. Lecture 21's frame law on overlaps is

$$e_j = e_i g_{ij}, \qquad A_j = A_i + g_{ij}^{-1}\mathrm{d}g_{ij}, \qquad g_{ij} \colon U_i \cap U_j \to \mathrm{U}(1).$$

Locally write $g_{ij} = e^{\mathrm{i}\chi}$. Then $\mathrm{d}(g_{ij}^{-1}\mathrm{d}g_{ij}) = \mathrm{d}(\mathrm{i}\mathrm{d}\chi) = 0$. Thus the F_i agree and define a global imaginary two-form F. It is closed because $\mathrm{d}F_i = \mathrm{d}^2 A_i = 0$.

Definition 22.1. For a smooth map $f \colon N \to M$, the *pullback bundle* $f^*E \to N$ has fibre $E_{f(y)}$ at $y \in N$.

Pull back the product charts to $f^{-1}(U_i)$. Their transitions $g_{ij} \circ f$ satisfy the same cocycle, so Lecture 20 makes this a smooth bundle.

Definition 22.2. A *characteristic class* assigns a cohomology class to a bundle, invariant under bundle isomorphism and compatible with pullback.

Definition 22.3. Here the *real first Chern class* is

$$c_1(E) = \left[\frac{\mathrm{i}}{2\pi}F\right] \in H^2_{\mathrm{dR}}(M;\mathbb{R}).$$

The form inside the brackets is real since F is imaginary. We use de Rham classes throughout this lecture. Torsion classes, those killed by a nonzero integer, are not detected by these real forms. The integer sphere computation below needs no torsion comparison.

Theorem 22.4 (Chern–Weil · *cited; line-bundle first Chern class proved given partitions of unity*)**.** *Every complex line bundle has a Hermitian metric and a unitary connection. The class $c_1(E)$ is independent of both choices and is a characteristic class. More generally, on a unitary rank-r bundle, the forms $\operatorname{tr}(F^k)$ are closed and their classes in $H^{2k}_{\mathrm{dR}}(M;\mathbb{C})$ are independent of the connection, for every integer $k \geq 1$. Here powers use matrix wedge multiplication. This general assertion is cited.*

Proof for the line-bundle class. First fix a Hermitian metric and compare two unitary connections. In the same local unit frames, both potentials change by the same transition term. Their differences therefore glue to a global imaginary one-form B. Their curvatures satisfy

$$F^{(1)} - F^{(0)} = \mathrm{d}B, \qquad \frac{\mathrm{i}}{2\pi}(F^{(1)} - F^{(0)}) = \mathrm{d}\left(\frac{\mathrm{i}}{2\pi}B\right).$$

Thus changing the connection leaves the real cohomology class unchanged.

Next we construct the metric and a unitary connection. Lecture 12's cited partitions of unity supply the auxiliary choices. Take a locally finite partition ψ_i subordinate to product charts U_i. For $v, w \in E_x$, the weighted sum of their standard chartwise Hermitian products is

$$\langle v, w\rangle = \sum_i \psi_i(x)\langle v, w\rangle_i.$$

It is positive definite since the nonnegative weights sum to one. Their supports and local finiteness make the sum smooth. Normalize the local frames for this metric. In each unit frame let $\nabla^{(i)}$ be the connection with potential zero. The sum $\nabla s = \sum_i \psi_i \nabla^{(i)} s$ defines a global connection. Each summand extends smoothly by zero. The sum is locally finite. The Leibniz and metric product rules are affine in the connection, so their weighted sums give the same rules because $\sum_i \psi_i = 1$. Thus this connection is unitary.

Finally we compare connections unitary for possibly different metrics. Use the same complex frames for both, without requiring them to be unitary. The line frame law also holds for $h \in \mathbb{C}^\times$. Locally write $h = e^{b+\mathrm{i}\chi}$ with b and χ real. Then $h^{-1}\mathrm{d}h = \mathrm{d}b + \mathrm{i}\mathrm{d}\chi$ is closed, so curvature is unchanged in any such frame. Write the two potentials as $A_i^{(0)}$ and $A_i^{(1)}$ in these common frames. Their difference $B_i = A_i^{(1)} - A_i^{(0)}$ has no transition derivative term, so $B_j = B_i$. These complex one-forms glue to a global B. Both curvatures are imaginary. Taking real parts in their exact difference gives

$$F^{(1)} - F^{(0)} = \mathrm{d}B, \qquad \frac{\mathrm{i}}{2\pi}F^{(1)} - \frac{\mathrm{i}}{2\pi}F^{(0)} = \mathrm{d}\operatorname{Re}\left(\frac{\mathrm{i}}{2\pi}B\right).$$

Their real de Rham classes coincide. For a smooth map $f\colon N \to M$, the pullback bundle has pulled-back local potentials. Its curvature is f^*F because pullback commutes with d. Thus $c_1(f^*E) = f^*c_1(E)$. An arbitrary complex bundle isomorphism transports the metric and connection together. Independence of both choices then proves invariance even if the isomorphism does not preserve a previously chosen target metric. ∎

The general statement is Nakahara, Theorem 11.1 and (11.20)–(11.21). No Riemannian metric or orientation on the base entered this definition.

Definition 22.5. For a closed oriented surface Σ, the bundle's *first Chern number* is

$$C_1(E;\Sigma) = \frac{\mathrm{i}}{2\pi}\int_\Sigma F.$$

Changing the surface orientation reverses this number. Stokes makes it independent of the representative curvature in the same class.

22.2 Integrality on the sphere

Give S^2 its outward orientation, equal to the complex orientation of $\mathbb{C}\mathrm{P}^1$. Let D_N and D_S be the closed northern and southern hemispheres. Their common equator C has increasing azimuth ϕ as ∂D_N. The southern boundary is $-C$.

Lemma 22.6 (Cap frames · *proved*)**.** *A Hermitian line bundle with a unitary connection has a smooth unit frame on each pole-deleted chart of the sphere.*

Proof. Identify the sphere minus the opposite pole with $\mathbb{R}^2$. Choose one unit vector in the fibre over its coordinate origin and transport it along every radial segment to its endpoint. Lecture 21 gives unique unitary transport. Smooth dependence of linear ODE solutions on the endpoint gives a smooth frame. Locally in the endpoint, a finite chain of trivializations covers these compact segments, so this smooth-dependence argument also applies when no initial global frame is given. Each resulting chart contains its entire closed hemisphere. ∎

Theorem 22.7 (The sphere Chern number · *proved*)**.** *Let $E \to S^2$ be a Hermitian line bundle with a unitary connection. Choose smooth unitary frames over neighbourhoods of D_N and D_S. In the frame order $e_S = e_N g_{NS}$,*

$$C_1(E;S^2) = \frac{\mathrm{i}}{2\pi}\int_{S^2} F = \frac{1}{2\pi\mathrm{i}}\oint_C g_{NS}^{-1}\mathrm{d}g_{NS} = \operatorname{wind}(g_{NS}) \in \mathbb{Z}.$$

The figure represents the hemispheres as oriented disks.

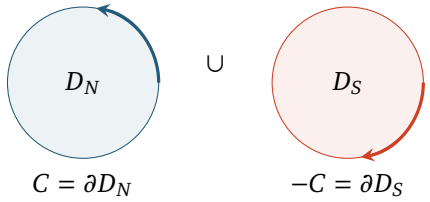


Figure 22.1: The two hemispheres induce opposite orientations on their common equator. Adding their Stokes formulas leaves $\oint_C (A_N - A_S)$, with both disks viewed from above the north pole.

Proof. In these frames, Stokes and the boundary orientations give

$$\frac{\mathrm{i}}{2\pi}\int_{S^2} F = \frac{\mathrm{i}}{2\pi}\left(\int_{D_N} \mathrm{d}A_N + \int_{D_S} \mathrm{d}A_S\right) = \frac{\mathrm{i}}{2\pi}\oint_C (A_N - A_S) = \frac{1}{2\pi\mathrm{i}}\oint_C g_{NS}^{-1}\mathrm{d}g_{NS}.$$

The last equality uses $A_S = A_N + g_{NS}^{-1}\mathrm{d}g_{NS}$ and $-\mathrm{i} = 1/\mathrm{i}$. Lecture 3's circle lifting theorem identifies the last integral with the integer winding. This proves the result for every such bundle, rather than only the monopole potentials. ∎

A different cap frame changes the transition by phases extending over the disks. Each such boundary phase has winding zero, so the integer is unchanged. Lecture 20's cited topological clutching classification applies to the principal $\mathrm{U}(1)$ bundle of unitary frames. It adds that equal integers give continuously isomorphic underlying bundles and that every integer occurs. The tautological line has transition $e^{-\mathrm{i}\phi}$, hence Chern number -1. The phase of the southern frame relative to the northern frame makes one or two positive turns in the following examples.

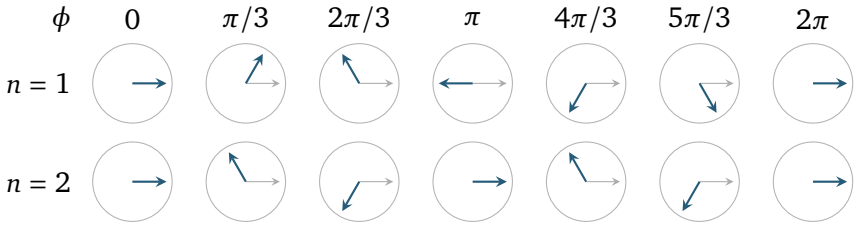


Figure 22.2: At each equatorial azimuth, the grey arrow represents e_N and the coloured arrow represents e_S relative to e_N. The transitions $e^{\mathrm{i}\phi}$ and $e^{2\mathrm{i}\phi}$ have windings +1 and +2.

22.3 The Dirac monopole

Definition 22.8. A *magnetic monopole* is a point source with nonzero outward magnetic flux through an enclosing sphere.

Let $q \neq 0$ be the signed particle charge and Φ_{m} the magnetic flux.

Physical input. We use the radial field model of Lecture 15 on the complement of the source. We describe a charged particle by local wave-function coefficients with the minimally coupled derivative $D = \mathrm{d} - (\mathrm{i}q/\hbar)A_{\mathrm{em}}$ of Lecture 21. Coefficients in overlapping frames describe one state and obey $\psi_S = g_{NS}^{-1}\psi_N$ for a single-valued phase transition. In the coherent-path model of Lecture 21, the electromagnetic phase along a path is the connection's transport phase. On a sphere enclosing the monopole, the real electromagnetic curvature is

$$F_{\mathrm{em}} = \frac{\Phi_{\mathrm{m}}}{4\pi}\sin\theta\,\mathrm{d}\theta\wedge\mathrm{d}\phi, \qquad \int_{S^2} F_{\mathrm{em}} = \Phi_{\mathrm{m}}.$$

For nonzero flux, Lecture 15's Stokes argument excludes a global potential on the sphere. The local coefficients therefore form a section of a line bundle rather than a function in a global frame. The *Wu–Yang potentials* on the two pole-deleted charts are

$$A_{\mathrm{em}}^{(N)} = \frac{\Phi_{\mathrm{m}}}{4\pi}(1-\cos\theta)\mathrm{d}\phi, \quad A_{\mathrm{em}}^{(S)} = -\frac{\Phi_{\mathrm{m}}}{4\pi}(1+\cos\theta)\mathrm{d}\phi,$$
$$A_N = -\frac{\mathrm{i}n}{2}(1-\cos\theta)\mathrm{d}\phi, \qquad A_S = +\frac{\mathrm{i}n}{2}(1+\cos\theta)\mathrm{d}\phi,$$

where $n = q\Phi_{\mathrm{m}}/(2\pi\hbar)$ is initially a real parameter. The conversion is $A = -(\mathrm{i}q/\hbar)A_{\mathrm{em}}$ and $F = -(\mathrm{i}q/\hbar)F_{\mathrm{em}}$. The imaginary connection potential includes $q/\hbar$ and is distinct from the real potential.

The polar coordinate ϕ fails at the poles, but these potentials are smooth on their claimed charts. For ambient unit-sphere coordinates (x, y, z),

$$(1 - \cos\theta)\mathrm{d}\phi = \frac{x\mathrm{d}y - y\mathrm{d}x}{1+z}, \qquad (1 + \cos\theta)\mathrm{d}\phi = \frac{x\mathrm{d}y - y\mathrm{d}x}{1-z}.$$

The first denominator is nonzero off the south pole, the second off the north pole. Both give the same smooth curvature $F = -(\mathrm{i}n/2)\sin\theta\mathrm{d}\theta \wedge \mathrm{d}\phi$.

The forms $(1 - \cos\theta)\mathrm{d}\phi$ and $(1 + \cos\theta)\mathrm{d}\phi$ equal $-4\pi a_U$ and $4\pi a_V$, respectively, for Lecture 13's cap primitives. Here U is the north chart and V is the south chart. Their rescalings satisfy

$$A_N = 2\pi\mathrm{i}n\, a_U, \qquad A_S = 2\pi\mathrm{i}n\, a_V, \qquad a_V - a_U = \frac{\mathrm{d}\phi}{2\pi}.$$

The de Rham calculation allowed every real multiple of the sphere class. For a charged bundle, the rescaled overlap difference must come from a single-valued phase transition.

Corollary 22.9 (Dirac quantization · *proved*)**.** *The Wu–Yang potentials define a unitary line-bundle connection for charge $q \neq 0$ exactly when*

$$q\Phi_{\mathrm{m}} = 2\pi\hbar n, \qquad n \in \mathbb{Z}.$$

Its sphere Chern number is n.

Proof. On the overlap, $A_S - A_N = \mathrm{i}n\mathrm{d}\phi$. The frame transition must therefore solve $g_{NS}^{-1}\mathrm{d}g_{NS} = \mathrm{i}n\mathrm{d}\phi$. It has the form $ce^{\mathrm{i}n\phi}$ for a constant $c \in \mathrm{U}(1)$. It is single-valued around the equator exactly when $e^{2\pi\mathrm{i}n} = 1$, which holds exactly for $n \in \mathbb{Z}$. For every such integer choose $g_{NS} = e^{\mathrm{i}n\phi}$ on the full overlap. This phase is smooth away from the poles. Lecture 20 reconstructs its smooth bundle. The connection law glues the two potentials. The sphere theorem gives the bundle's Chern number n. ∎

This is a conditional charge-quantization argument, not an assertion that monopoles exist. The potentials follow Nakahara, §10.5.2, with its imaginary-potential and north/south transition choices translated by the displayed conversion and frame order. For a fixed nonzero monopole flux, every allowed electric charge must therefore belong to $(2\pi\hbar/\Phi_{\mathrm{m}})\mathbb{Z}$.

Definition 22.10. The *Dirac string* is the singular negative z-axis of the radially extended northern potential in the one-potential description.

On a latitude traversed with increasing ϕ, its circulation is

$$\oint_{\theta=\text{constant}} A_{\mathrm{em}}^{(N)} = \frac{\Phi_{\mathrm{m}}}{2}(1 - \cos\theta) \longrightarrow \Phi_{\mathrm{m}} \quad \text{as } \theta \to \pi.$$

Lecture 21's holonomy formula therefore gives the limiting phase $e^{\mathrm{i}q\Phi_{\mathrm{m}}/\hbar}$ around the string. The string is invisible to this charged particle exactly when this phase is 1. This is the same quantization condition. The string is a gauge singularity, not an additional physical solenoid. For positive flux, the string flux in this singular description points toward the origin along the negative z-axis. The figure distinguishes this direction from the signed cap circulations.

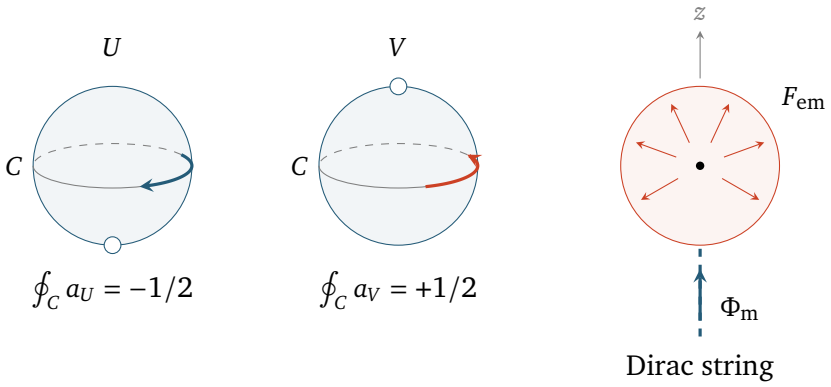


Figure 22.3: The cap primitives have opposite circulation signs along increasing azimuth. For positive Φ_{m}, the northern-potential description has a string on the negative z-axis with flux directed toward the origin.

A trial half-integer parameter gives smooth local potentials but fails the single-valued transition condition, as the phase paths show.

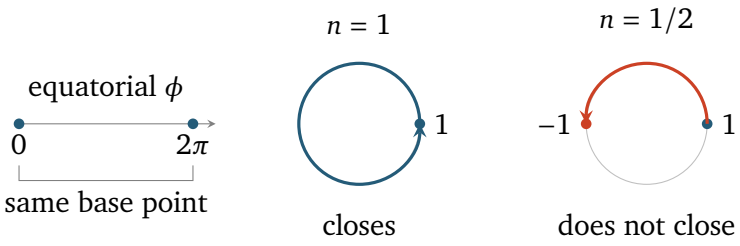


Figure 22.4: As ϕ runs from 0 to 2π, the phase $e^{\mathrm{i}n\phi}$ closes for $n = 1$. For the trial value $n = 1/2$, its final value is -1 instead of 1, so it cannot define a transition on the equator.

For $n = -1$, the monopole line is isomorphic to Lecture 20's tautological line. Both have transition $e^{-\mathrm{i}\phi}$ in the stated frame order. The principal circle bundle of its unit frames is the Hopf bundle $S^3 \to \mathbb{C}\mathrm{P}^1$. A unit frame is a unit vector in the tautological fibre, hence a point of S^3.

The cap integral need not be an integer. For a northern cap of polar radius $\alpha \in [0, \pi]$, the same curvature gives

$$\frac{\mathrm{i}}{2\pi} \int_{D_\alpha} F = \frac{n}{2}(1 - \cos\alpha).$$

The figure plots $n = 1$. Its hemisphere value is $1/2$, although the sphere number is 1.

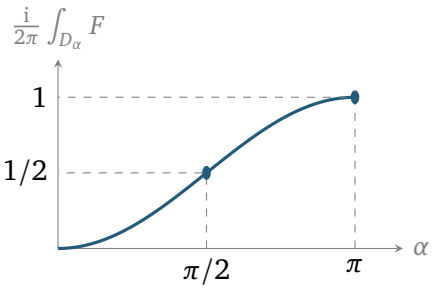


Figure 22.5: For the charge-one curvature, cap flux varies continuously with cap size. Only the full closed-sphere integral is constrained to the integer 1.

22.4 The tangent sphere as a complex line

On an oriented Riemannian surface Σ, let J rotate a positive orthonormal frame (e_1, e_2) by $Je_1 = e_2$ and $Je_2 = -e_1$. Positive orthonormal frame changes commute with this rotation, so it defines multiplication by i on every tangent plane. Its Hermitian fibre product is $\langle u, v\rangle = g(u, v) + \mathrm{i}g(Ju, v)$ for $p \in \Sigma$ and $u, v \in T_p\Sigma$. Lecture 17's Levi-Civita connection has $\nabla e_1 = -\omega^1{}_2 e_2$ and $\mathrm{d}\omega^1{}_2 = K\mathrm{d}\mathcal{A}$, where $\mathrm{d}\mathcal{A}$ denotes the oriented area form. This area symbol is distinct from the exterior derivative $\mathrm{d}A$ of a connection potential. The Levi-Civita connection's skew matrix commutes with J, hence the connection is complex-linear and unitary on this complex line.

Proposition 22.11 (The tangent Chern number · *proved*). *For a closed oriented Riemannian surface Σ, its tangent complex line has*

$$A = -\mathrm{i}\omega^1{}_2, \qquad F = -\mathrm{i}K\mathrm{d}\mathcal{A}, \qquad C_1(T\Sigma; \Sigma) = \frac{1}{2\pi}\int_\Sigma K\mathrm{d}\mathcal{A} = \chi(\Sigma).$$

In particular, $C_1(TS^2; S^2) = 2$.

Proof. The complex frame is e_1, with $e_2 = \mathrm{i}e_1$. Thus $\nabla e_1 = -\mathrm{i}\omega^1{}_2 e_1$, proving the connection formula. Differentiate and use Lecture 17's surface structure equation to obtain the curvature. Multiplication by $\mathrm{i}/(2\pi)$ gives $K\mathrm{d}\mathcal{A}/(2\pi)$. Lecture 18's Gauss–Bonnet theorem evaluates its integral as $\chi(\Sigma)$. For the sphere the Euler characteristic is 2, as required. ∎

This agrees with Lecture 20's tangent-frame winding +2. The inverse component transition winds −2. It is not the transition used in the Chern-number theorem.

22.5 Higher Chern classes

For higher-rank bundles, curvature matrices change by conjugation. Invariant polynomials give global scalar forms from these local matrices. The degree-four form gives the second Chern number used for nonabelian fields in Lecture 23. For a unitary rank-r bundle, use the defining $r \times r$ matrix curvature F. The frame law $F' = h^{-1}Fh$ and conjugation invariance of the determinant make the following form global. Even-degree forms commute as scalar entries.

Definition 22.12. The *total Chern form* is defined by

$$c(F) = \det\left(I + \frac{\mathrm{i}F}{2\pi}\right) = 1 + c_1(F) + c_2(F) + \cdots + c_r(F).$$

The component $c_j(F)$ has degree $2j$ and vanishes when $2j > \dim M$. The corresponding cohomology classes are the *Chern classes*.

The invariant-polynomial form of Chern–Weil, Nakahara Theorem 11.1, gives their closedness and connection independence. It applies to each determinant coefficient, not only the traces of powers. For two connections, their difference $B = A^{(1)} - A^{(0)}$ is an endomorphism-valued one-form. With $A_t = A^{(0)} + tB$ and curvature F_t, the cited relative transgression expresses the difference as an exterior derivative:

$$\operatorname{tr}\big((F^{(1)})^k\big) - \operatorname{tr}\big((F^{(0)})^k\big) = \mathrm{d}\left(k\int_0^1 \operatorname{tr}(B \wedge F_t^{k-1})\mathrm{d}t\right).$$

The trace expression inside is global, although the individual potentials are local. Nakahara's (11.20)–(11.21) supplies this assertion. Expanding the determinant through degree four gives

$$c_1(F) = \frac{\mathrm{i}}{2\pi}\operatorname{tr} F, \qquad c_2(F) = \frac{\operatorname{tr}(F\wedge F) - \operatorname{tr} F \wedge \operatorname{tr} F}{8\pi^2}.$$

This uses the elementary determinant coefficient $((\operatorname{tr} X)^2 - \operatorname{tr}(X^2))/2$ with $X = \mathrm{i}F/(2\pi)$. For an $\mathrm{SU}(r)$ connection, $\operatorname{tr} F = 0$, so $c_2(F) = \operatorname{tr}(F\wedge F)/(8\pi^2)$. Lecture 23 uses this normalization for its second Chern number.

Example 22.13 (A rank-two Chern form)**.** On the trivial rank-two bundle over $\mathbb{R}^4$, take the diagonal unitary potential

$$A = \operatorname{diag}(-2\pi\mathrm{i}x^1\mathrm{d}x^2, -2\pi\mathrm{i}x^3\mathrm{d}x^4).$$

Each diagonal one-form wedges with itself to zero, so $A \wedge A = 0$. Its curvature and total Chern form are

$$F = \operatorname{diag}(-2\pi\mathrm{i}\mathrm{d}x^1\wedge\mathrm{d}x^2, -2\pi\mathrm{i}\mathrm{d}x^3\wedge\mathrm{d}x^4),$$
$$c(F) = (1 + \mathrm{d}x^1\wedge\mathrm{d}x^2)(1 + \mathrm{d}x^3\wedge\mathrm{d}x^4).$$

The degree-four term is $c_2(F) = \mathrm{d}x^1\wedge\mathrm{d}x^2\wedge\mathrm{d}x^3\wedge\mathrm{d}x^4$. It equals $\mathrm{d}(x^1\mathrm{d}x^2\wedge\mathrm{d}x^3\wedge\mathrm{d}x^4)$, so its cohomology class is zero, as required for this trivial bundle.

Self-check. Show that a trivial Hermitian line bundle over a closed oriented surface has Chern number zero. Deduce what a nonzero Chern number says about the existence of a global unit frame.

Answer. Choose a global unit frame, so the connection has a global one-form A and $F = \mathrm{d}A$. Stokes gives $\int_\Sigma F = \int_{\partial\Sigma} A = 0$ because $\partial\Sigma$ is empty. A nonzero Chern number therefore excludes a global unit frame.

Self-check. A unitary line-bundle connection has cap potentials whose restrictions to the positive equator $C = \partial D_N$ are

$$A_N|_C = -\mathrm{i}(2+\cos\phi)\mathrm{d}\phi, \qquad A_S|_C = \mathrm{i}(1+\cos\phi)\mathrm{d}\phi.$$

Find one compatible equatorial frame transition and the sphere Chern number. Compute both hemisphere contributions. Explain why swapping the cap order leaves the sphere number unchanged. Decide whether $h = e^{\mathrm{i}\phi}$ is a smooth frame change on the whole northern chart.

Answer. The difference is $A_S - A_N = \mathrm{i}(3+2\cos\phi)\mathrm{d}\phi$. Thus $g_{NS} = e^{\mathrm{i}(3\phi+2\sin\phi)}$ satisfies the frame law. Its lifted phase increases by 6π, so the sphere number is 3. Stokes gives the two cap contributions

$$\frac{\mathrm{i}}{2\pi}\int_{D_N} F = 2, \qquad \frac{\mathrm{i}}{2\pi}\int_{D_S} F = 1.$$

Swapping caps inverts the transition and reverses the equator orientation. These two sign changes leave the number 3 unchanged. The proposed frame change has limit 1 along $\phi = 0$ and -1 along $\phi = \pi$ at the north pole, so it does not extend there.

Self-check. State in one sentence what the first Chern class is and why it was introduced. Explain why a curvature integral over a cap does not have the same integrality guarantee.

Answer. The first Chern class is the real cohomology class $[\mathrm{i}F/(2\pi)]$ of a line bundle, introduced to retain information about the bundle when its connection changes. For a fixed metric, two connections give curvatures differing by $\mathrm{d}B$ for a global imaginary one-form B, so the classes coincide. The sphere integrality proof uses a closed equator to measure transition winding. A cap has a boundary, and its integral alone measures no such winding. For $n = 1$, the hemisphere number is $1/2$.

Unit 22B Holomorphic monopole lines and their sections

22.6 The degree-n holomorphic line

We express the monopole line in frames where a covariant differential equation becomes ordinary holomorphicity of the coefficients. Global holomorphic sections must also satisfy the transition law at the opposite pole. We construct two such sections of $\mathcal{O}(1)$ below. Lecture 28 uses the same frames to turn a zero-mode equation into a count of sections.

Definition 22.14. In a complex chart $z = x + \mathrm{i}y$, a smooth function is holomorphic if $\partial_{\bar z} f = 0$, where $\partial_{\bar z} = \frac{1}{2}(\partial_x + \mathrm{i}\partial_y)$.

Definition 22.15. A *holomorphic line bundle* has local frames with holomorphic transition functions.

Definition 22.16. The $(0,1)$ *part* of a complex one-form is its $\mathrm{d}\bar z$ term. Write $\bar\partial f = (\partial_{\bar z} f)\mathrm{d}\bar z$.

The complex differentials satisfy

$$\mathrm{d}z = \mathrm{d}x + \mathrm{i}\mathrm{d}y, \qquad \mathrm{d}\bar z = \mathrm{d}x - \mathrm{i}\mathrm{d}y, \qquad \mathrm{d}f = (\partial_z f)\mathrm{d}z + (\partial_{\bar z} f)\mathrm{d}\bar z,$$

where $\partial_z = \frac{1}{2}(\partial_x - \mathrm{i}\partial_y)$. Use the projective coordinates z and $w = 1/z$ of Lecture 20. Here z denotes the complex coordinate rather than the ambient height used in §22.3.

Definition 22.17. Define the *degree* of a line over $\mathbb{C}\mathrm{P}^1$ to be its Chern number in the complex orientation. Define the *degree-n line* $O(n)$ for $n \in \mathbb{Z}$ by holomorphic frames with

$$e_S^{\mathrm{hol}} = e_N^{\mathrm{hol}} z^n.$$

Definition 22.18. A local section is *holomorphic* exactly when its holomorphic-frame coefficient satisfies $\partial_{\bar{z}} f = 0$. A global holomorphic section has such coefficients on every chart, obeying the section transition law.

Example 22.19 (Two global sections of $O(1)$)**.** Since $e_S^{\mathrm{hol}} = e_N^{\mathrm{hol}} z$, a section has coefficients related by $f_S(w) = w f_N(1/w)$. The northern coefficient $f_N(z) = 1$ therefore gives $f_S(w) = w$. It defines a global holomorphic section s_0, with its only zero at the south pole. The northern coefficient $f_N(z) = z$ gives $f_S(w) = 1$. This section s_1 has its only zero at the north pole. Their linear combinations have coefficients

$$f_N(z) = a + bz, \qquad f_S(w) = aw + b, \qquad a, b \in \mathbb{C}.$$

Both coefficients are holomorphic on their whole charts and satisfy the transition law. Evaluation at $z = 0$ and $z = 1$ shows that s_0 and s_1 are linearly independent.

The diagram shows the relation between holomorphic and unitary frames.

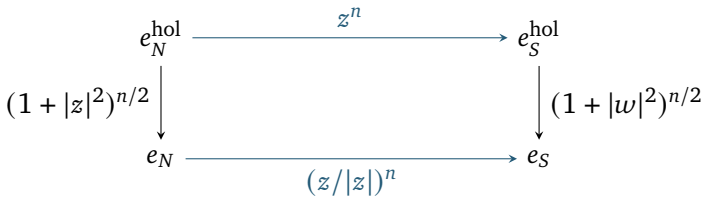


Figure 22.6: Positive real rescalings turn holomorphic frames into unitary frames. Their transition is the phase $(z/|z|)^n$, with winding n.

Example 22.20 (The monopole line as $O(n)$)**.** Identify the Wu–Yang monopole line with $O(n)$, for $n \in \mathbb{Z}$. Equip these frames with squared norms $\|e_N^{\mathrm{hol}}\|^2 = (1+|z|^2)^{-n}$ and $\|e_S^{\mathrm{hol}}\|^2 = (1+|w|^2)^{-n}$. They agree on the overlap since $|w| = 1/|z|$ and $e_S^{\mathrm{hol}} = e_N^{\mathrm{hol}} z^n$. The unitary frames are

$$e_N = (1+|z|^2)^{n/2} e_N^{\mathrm{hol}}, \qquad e_S = (1+|w|^2)^{n/2} e_S^{\mathrm{hol}}.$$

Their transition is $(z/|z|)^n$. In the northern unit frame the Wu–Yang connection is

$$A_N = -\frac{n}{2} \frac{\bar{z}\mathrm{d}z - z\mathrm{d}\bar{z}}{1+|z|^2}.$$

Changing to $e_N^{\mathrm{hol}} = (1+|z|^2)^{-n/2} e_N$ gives

$$A_N^{\mathrm{hol}} = A_N - \frac{n}{2}\mathrm{d}\log(1+|z|^2) = -\frac{n\bar{z}}{1+|z|^2}\mathrm{d}z.$$

In the southern holomorphic frame the same calculation gives

$$A_S^{\mathrm{hol}} = -\frac{n\bar{w}}{1+|w|^2}\mathrm{d}w = A_N^{\mathrm{hol}} + n\frac{\mathrm{d}z}{z} \quad \text{on the overlap.}$$

Both expressions have no antiholomorphic differential term and are smooth at their own chart origin. Thus the $(0,1)$ part of the covariant derivative is ordinary $\bar\partial$ in these frames. This equips the monopole line with the displayed holomorphic structure. Its Chern number is n by the equatorial winding theorem. For the coefficient ψ in the unit frame the same equation is $(\partial_{\bar z} + nz/[2(1+|z|^2)])\psi = 0$. This local equation will reappear in Lecture 28 with its stated spinor twist.

With the displayed metric for $O(1)$, the two sections have norms $\|s_0\| = \cos(\theta/2)$ and $\|s_1\| = \sin(\theta/2)$. The figure shows these lengths in separate fibres above points of the meridian $\phi = 0$.

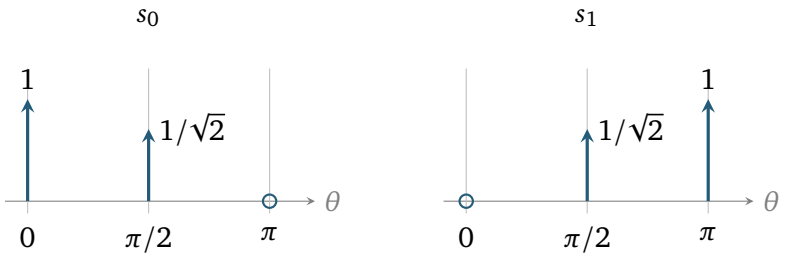


Figure 22.7: The two sections of $O(1)$ vanish at opposite poles. Vertical arrows lie in the line fibres, with lengths given by the displayed norms.

The cited holomorphic classification on $\mathbb{C}\mathrm{P}^1$ says that every holomorphic line is isomorphic to exactly one $O(n)$, with n its degree. See Gunning, *Some Topics in the Function Theory of Compact Riemann Surfaces*, preliminary version (2018), Theorem 2.2, (2.3) and Corollary 2.3, pp. 34–35. This is a holomorphic result in addition to Lecture 20's continuous clutching classification. The tautological frames $(1, z)^T$ and $(w, 1)^T$ have holomorphic transition z^{-1}. The tangent coordinate frames have holomorphic transition $-z^2$ by Lecture 20. Their degrees and the cited classification identify the tautological line with $O(-1)$ and the tangent line with $O(2)$.

Summary. A line bundle's real first Chern class is $[\mathrm{i}F/(2\pi)]$ and is independent of the connection. On the outward sphere its integral is the winding of the frame transition $e_S = e_N g_{NS}$ along the northern boundary. The monopole condition is $q\Phi_\mathrm{m} = 2\pi\hbar n$ with $n \in \mathbb{Z}$. The tangent sphere has Chern number 2, while $O(n)$ has number n. A given complex bundle's class needs no base metric or orientation. The tangent example uses an oriented metric to define its complex structure. Signed surface integrals need the orientation. Lecture 23 uses the second Chern form to study nonabelian fields.

Reading. Nakahara, §10.5.2 and §11.1–11.2. Tu, *Differential Geometry*, §25.8, p. 235. Gunning, Theorem 2.2, (2.3) and Corollary 2.3, pp. 34–35, for the holomorphic classification.